

Abiyaz Chowdhury

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EDUCATION

- **Stony Brook University** Stony Brook, NY
Doctor of Philosophy in Computer Science (GPA: 3.8) May 2027
- **Cooper Union for the Advancement of Science and Art** New York, NY
Bachelor of Engineering in Electrical Engineering

EXPERIENCE

- **Stony Brook University** Stony Brook, NY
Research Assistant
 - Engineered **GRAPHZEPPELIN**, a high-performance C++ streaming graph processing system, achieving 2-5 million edge updates/second on dynamic, dense graphs too large for RAM.
 - Designed and implemented **CUBESKETCH**, a novel l_0 sketching data structure, reducing space by 4x and increasing update speed by 3 orders of magnitude compared to state-of-the-art for graph connectivity.
 - Optimized I/O efficiency and enabled out-of-core computation via node-based buffering and gutter trees, allowing processing of graphs 2 orders of magnitude larger on SSD with only 29% performance degradation.
 - Implemented multi-threaded stream ingestion, scaling throughput by 25x with 46 threads using OpenMP.
 - Conducted empirical research on cache-oblivious algorithms (C++) for matrix multiplication and sorting, demonstrating their practical adaptivity to dynamic memory fluctuations in multi-program environments.
 - Developed a novel memory profile generator by applying memory ballooning techniques to Linux cgroups to simulate adversarial and realistic memory workloads, enabling rigorous performance evaluation of algorithms .
 - Analyzed I/O performance, revealing that cache-oblivious designs (e.g., Lazy Funnel Sort) consistently outperformed cache-aware alternatives (e.g., External-memory Merge Sort) by up to 7.1x in disk I/Os under realistic memory fluctuations.
- **Stony Brook University** Stony Brook, NY
Teaching Assistant
 - TA for 9 CS courses (30–105 students), including graduate NLP, Algorithms, and Software Engineering.
 - Mentored junior TAs and collaborated with faculty to deliver high-quality course content.
 - Designed assignments, exams, lecture slides, and automated grading systems using Linux tools (e.g., **cgroups**, **ulimit**) to sandbox student code.
 - Integrated **Moss** for plagiarism detection, reviewed reports, and handled academic integrity cases
 - Delivered recitations and substitute lectures on concurrency, graph theory, recursion, and related topics.
- **Florida Institute for Human and Machine Cognition (IHMC)** New Orleans, LA
Engineering Intern
 - Built and integrated neural networks for semantic role labeling in **Saul**, a declarative ML/NLP framework built on Java and Scala.

SKILLS

- **Languages:** Scala, Python, C, C++, SQL, Java, Bash
Technologies: AWS, Kubernetes, Linux, Git, Docker, CI/CD
Systems: Distributed Systems, High-Performance Computing (HPC), I/O Optimization, Memory Management, Multithreading, Linux Environment